



1. I can use the proportional relationship between the measures of segments formed by intersecting chords to solve problems.



Learning Target

- I can solve mathematical and real-world problems that involve the lengths formed by the intersection of two secants.





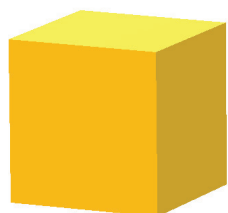
12.2.1 Warm-Up: Would You Rather?

1. Would you rather have ...

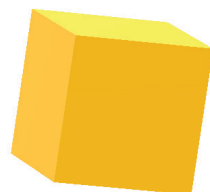
A cube of gold, 25 millimeters(mm) on each side ...

OR

Two cubes of gold, one is 24 mm per side, one is 1 mm per side ...



25 mm



24 mm



1 mm

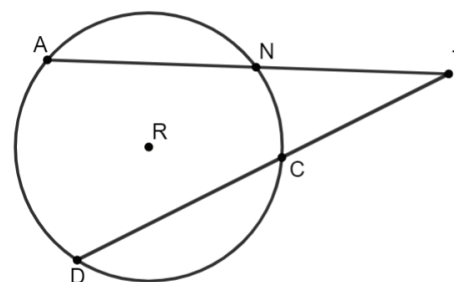
2. Explain your choice.



12.2.2 Exploration: Intersecting Secant Lengths

A **secant** is a line or line segment that passes through two points on a circle. In the interior of the circle, a secant line segment forms a chord.

1. Circle R is shown with points A , N , D , and C on the circle. Secants $\overline{AT}$ and $\overline{DT}$ intersect at point T outside the circle. $AN = 6$, $CD = 9$, $NT = 8$, and $CT = 7$.



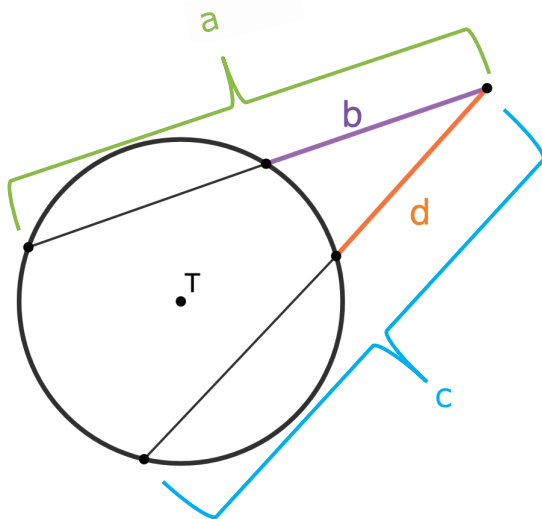
a. Complete the table.

Product	Work	Answer
$AN \cdot NT$		
$AN \cdot AT$		
$AT \cdot NT$		
$DC \cdot CT$		
$DC \cdot DT$		
$DT \cdot CT$		

b. Discuss the following with your partner.

- What do you notice about your answers?
- What do you notice about the secant segments for the rows that have the same answer?

c. With your partner, use the diagram of Circle T to write a formula that can be used to find the segment lengths of intersecting secants.

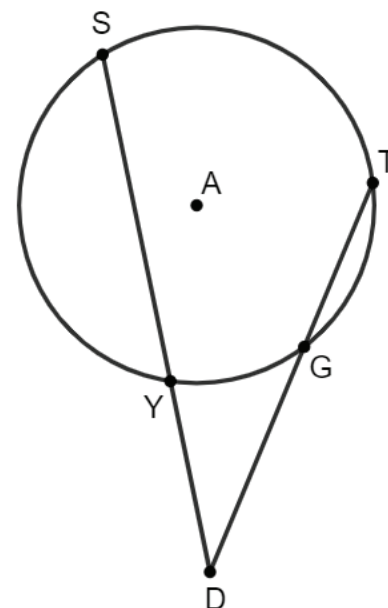


If two secants intersect outside a circle, then the product of the lengths of one secant segment and its external secant segment equals the product of the lengths of the other secant segment and its external secant segment.

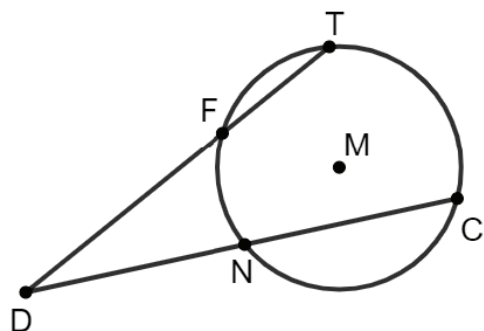


12.2.3 Guided Practice: Solving Problems Involving Intersecting Secant Lengths

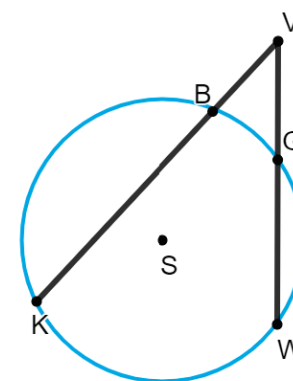
1. Circle A with secants $\overline{SD}$ and $\overline{TD}$ is shown. $\overline{SD}$ and $\overline{TD}$ intersect at point D outside the circle so that $SY = 5$, $DY = 3$, $TG = x$, and $DG = 4$. Solve for x .



2. Circle M is shown, where points T , F , N , and C are on circle M . Secants $\overline{TD}$ and $\overline{CD}$ intersect at point D outside the circle such that $TF = 7$, $FD = 9$, $CN = 5x$, and $ND = 4x$. Determine the value of x .



3. Leon Sinks Geologic Area is located in Apalachicola National Forest. The area has 18 sinkholes, many of which create a cave system. A park ranger mapped a circular sinkhole. His diagram is shown, where circle S has secants $\overline{VK}$ and $\overline{VW}$ such that points B and G are on circle S and point V is outside the circle. $KB = (4x + 4.25)$ meters (m), $VB = 12.75$ m, $GW = (3x + 5.5)$ m, and $GV = 14$ m. Find the distance, in meters, from point K to point B .

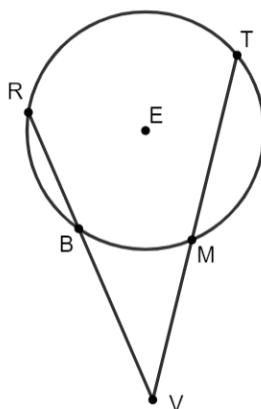




12.2.4 Your Turn!

- Circle E has points R , F , G , and M on the circle. Secants $\overline{RZ}$ and $\overline{MZ}$ intersect at point Z outside the circle. Point G is on $\overline{MZ}$, and point F is on $\overline{RZ}$. $MG = 10$, $GZ = 8$, $RF = 2x - 17$, and $FZ = 9$. Solve for x .

- Tammie was calculating the distance between her favorite places to visit in Epcot. Circle E represents the World Showcase where R , T , M , and B are on the circle and point V is outside the circle at Spaceship Earth. Tammie knows that $RB = 975$ feet (ft), $BV = 1425$ ft, and $VM = 1250$ ft. What is the length of $\overline{MT}$, in feet?

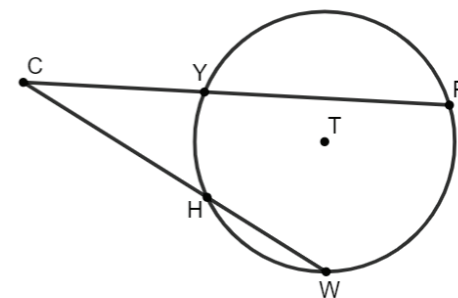


12.2.5 Wrap-Up: Error Analysis

- Donald was finding the value of x in circle T , where Y , P , W , and H are points on the circle. He knows that $\overline{CP}$ and $\overline{CW}$ are secants of the circle, where $CY = 4$, $YP = x - 3$, $CH = 5$, and $HW = x - 6$. His work is shown.

Donald's Work

$$\begin{aligned} 4(x - 3) &= 5(x - 6) \\ 4x - 12 &= 5x - 30 \\ x &= 18 \end{aligned}$$



Write a note to Donald explaining the mistake he made when solving for x .